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Chapter #5

Statistical methods

Definition of statistics:

⇒ Statistics is the science of:

- Collecting data
- Organizing data
- Drawing conclusions from data

⇒ There are two types of statistics:

1. Descriptive statistics:

⇒ Describes general characteristics of data.

⇒ Involves summarization and organization.

2. Statistical inference:

⇒ Methods of drawing conclusions from data.

⇒ Fundamental principles are emphasized.

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Statistical data analysis in data mining:

Significance in data mining:

⟹ Statistical data analysis is a well established methodology for data mining.

⟹ Early computer based data analysis applications were developed with statistician's support.

Range of data analysis:

⟹ Methods range from 1D data-analysis to multivariate data analysis.

⟹ Statistics offers various methods for data mining, including:

- Different types of regression analysis
- Discriminant analysis

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Statistical inference

Q. Explain methods of statistical inference commonly used in data mining?

Statistical inference:

⟹ Process of using data from a sample to draw conclusions about a population.

⟹ It involves making predictions, estimations and decisions about the population based on the information provided by the samples.

Population:

⟹ Def:

⟹ A population is the entire set of observations we are interested in analyzing.

Components:

⟹ Can include:

- People, objects, events, or anything of statistical interest.

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Size:
- Can be finite or infinite.

challenges with population:
- Observing the entire population is often impossible or impractical.

Example:
- Testing the lifespan of all lights bulbs of a certain brands would be impractical.

Sample:
Def:
⇒ Subset of population is called sample.

- Used to draw conclusions about the population.

Building models:
- For a data set (sample), a statistical method is created to make predictions or inferences about the population.
- Samples should be accurately represent the population to make valid inferences.

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Sampling bias:
Def:
- Bias occurs when a sampling procedure consistently overestimates or underestimates some characteristics of the population.

Avoid bias:
- To avoid bias, samples should be chosen randomly and independently.

Methods of SI / branches

⇒ There are two main categories of statistical inference:
- Estimation
- Hypothesis testing

1. Estimation:
Def:
⇒ Finding approximate values for population parameters based on sample data.

estimator: Rule or method that is used to estimate a parameter.

estimate: a numeric value obtained ~~from~~/by ~~substituting the sample~~ observations in the rule or formula.

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Goal:

- To estimate one or more parameters (unknown values) of the real world.

Types of estimation:

⇒ There are two types of estimation:

- Point estimation
- Interval estimation

Point estimation:

Def:

⇒ Provides a single value as an estimate of a population parameter.

ex:

- Estimate the mean(μ) of population using sample mean $\bar{x}$.

Interval estimation:

Def:

⇒ Provides a range of values within which the population parameter is expected to lie.

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ex: Estimating the average income with a confidence level/interval (e.g 40,000 to 50,000 with 95% confidence).

Goal of estimation:

- To gain information from a dataset T in order to estimate one or more parameters w belonging to the model of the real world system. $f(x, w)$.
- A dataset T is described by the ordered n-tuples of values for variables:

$$X = \{X_1, X_2, \ldots, X_n\}$$
$$T = \{(x_{11}, \ldots, x_{1n}), (x_{21}, \ldots, x_{2n}), \ldots, (x_{m1}, \ldots, x_{mn})\}$$

- it can be organized in a tabular form as a set of samples with its corresponding feature values.

Using estimates for predictions:

⇒ Once parameters are estimated the model can predict the value of a variable Y based on other variable $X^* = X - Y$

Regression: if Y is numeric

classification: if Y is discrete, unordered.

Prediction error:

- The difference between the predicted value of $f(X^*, w)$ and the actual value of Y.
- for quality measure we expected mean-squared error.

$$E_T[(Y - f[X^*, w])^2]$$

Sample $\bar{x}$:

- $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

Sample variance s^2:

- $s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

2. Hypothesis Testing:-

Hypothesis definition:

Statistical hypothesis:

⇒ An assertion or conjecture about one or more populations.

⇒ The truth or falsity of a statistical hypothesis can never be known with absolute certainty, unless we examine the entire population.

Null hypothesis (H_0):

- The hypothesis to be tested is called null hypothesis.

Alternative hypothesis:

- Accepted when H_0 is rejected.

Process:

(1) formulating hypothesis:

- Null hypothesis
- Alternative hypothesis

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(2) choosing a significance level:

○ Commonly denoted as α probability of rejecting the null hypothesis when H is true $\alpha = 0.05$.

(3) Calculating a test statistic:

○ A value computed from the sample data that is used to make decision about the hypothesis.

e.g z-score, t-score, chi-square

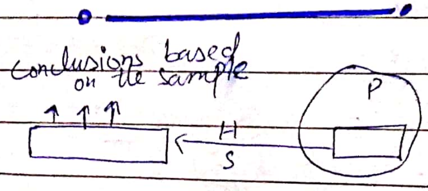
(4) making a decision:

P-value:

○ if the P-value (~~H_0~~) is less than α reject H_0.

Confidence interval:

○ if the CI does not contain the parameter value stated in H_0, reject H_0.



Topics:

Assesing differences in data sets:

Q- Identify different statistics parameters for assesing difference in data sets?

⇒ Assesing differences in data sets involves using various statistical parameters to understand and compare central tendency, dispersion and other characteristics of data.

⇒ Some of the key parameters:

1. Measure of central tendency:

Def:

⇒ Are statistical metrics that describe the location of the middle or center point of a data distribution.

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⇒ They provide insight into where most of the values of a given attribute falls.

⇒ The key measure of central tendency:

1. Mean (A.M):

Def:

⇒ The average value of all data points in a set.

formula:

$$\Rightarrow \frac{1}{n}\sum_{i=1}^{n} x_i$$

- x_i represent each data point
- n is the number of data point

Use:

- Useful for datasets without extreme values (outliers).

weighted mean:

-

$$W.M = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

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2. median:

Def:

⇒ The middle most value of an ordered data set.

for odd dataset:

- $\tilde{X} = x\left(\frac{n+1}{2}\right)^{th}$

for even dataset:

- $\tilde{X} = \dfrac{x\left(\frac{n}{2}\right) + x\left(\frac{n+1}{2}\right)}{2}$

Use:

⇒ for skewed data

⇒ for outliers data

⇒ no affected by extreme values

3. mode:

Def:

The most frequently occuring value in a data set.

formula & classification:

Unimode: one mode

multimode: more than one mode (e.g bimodal, trimodal)

Usecase:

- Useful for categorical data.
- Also used for numeric data

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4. Midrange:

Def:

⇒ The midrange is the value halfway between the maximum and minimum values in a data set.

formula:

- $\frac{max(x) + min(x)}{2}$

Usecase:

- Highly sensitive to outliers.
- Provide quick estimate of the center of the data.

2. Measure of dispersion

Def:

⇒ Statistical metrics that describes the spreadness or variability of a data points in a data set.

⇒ They indicate how much the data values deviate from the central tendency.

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1. Range:

Def:

⇒ Difference between the maximum and minimum values in a data set.

formula:

range = max(x) − min(x)

Use case:

⇒ Give a quick sense of the spread.

2. Variance (σ^2):

Def:

⇒ The average squared difference between each data point and the mean.

formula:

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - mean)^2$$

Use case:

- Provides insight into the data spread around the mean.

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3. Standard deviation (σ):

Def:

⇒ The square root of the variance

⇒ representing the average distance from the mean.

formula:

$$\sigma = \sqrt{\sigma^2}$$

Properties of SD:

- measure spread about the mean.
- Non-negative: $\sigma = 0$ if all data points are identical, otherwise $\sigma > 0$.

Use case:

- Useful when comparing the spread between different data sets.

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4. Interquartile range:

Def:

⇒ The range within which the middle 50% of the data point lie.

formula:

- $IQR = Q_3 - Q_1$

Q_3 → median of the upper half of the data.

Q_1 → median of the lower half of the data

Measure of Shape

1. Skewness:

Def:

⇒ A measure of the asymmetry of the distribution of values in a dataset.

formula:

$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - mean}{\sigma}\right)^3$$

Interpretation:

+ive skew: Tail on the right side

-ive skew: Tail on the leftside.

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2. Kurtosis:

Def:

A measure of the tailedness of the distribution of values in a data set.

formula:

$$\text{kurtosis} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left(\frac{x_i - \text{mean}}{\sigma}\right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

↑ kurtosis: more data in the tails.

↓ kurtosis: less data in the tails.

4. Measure of relationship

1. Correlation Coefficient (Pearson's r):

Def:

⟹ A measure of the linear relationship between two variables.

formula:

$$r = \frac{\sum_{i=1}^{n}(x_i-\bar{x}_i)(y_i-\bar{y}_i)}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x}_i)^2(y_i-\bar{y}_i)^2}}$$

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$r = 1$: Perfect +ive

$r = -1$: Perfect -ive

$r = 0$: No linear

2. Covariance:

Def:

⟹ A measure of how much two variables are changes together.

formula:

$$\circ\ COV(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$$

Interpretation:

+ive cov: Tend to increase together.

-ive cov: one variable ↑ when other ↓

Visualization Tools

1. Boxplot:

- A graphical representation showing the distribution of data through their quartiles.

Components: minimum, maximum, Q_1, median, Q_3.

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2. Histogram:

- A bar graph that shows the frequency distribution of a data set.

Use:

- Identifies the shape of the distribution.

3. Scatter Plot:

- A graph that shows the relationships between two variables.

Use:

- Visualize correlations & identifies patterns.

Five-number Summary:

⇒ Concise way to describe the distribution of a data set.

⇒ Provides a snapshot of the spread and center of the data.

⇒ minimum, Q_1, $\bar{x}$, Q_3, maximum.

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Inference

what we think about system? → what system think about us? → bayesian inference → based on system thinking → intelligent inference

Topic: Bayesian inference

Q. Describe the components of Naive Bayesian Classifier and basic principles?

→ supervised learning algorithm → based on bayes theorem. → Naive means all variables are independent

Bayesian inference:

⇒ Bayesian inference is a method that uses probability to incorporate external information into data analysis.

⇒ This approach is especially useful when the data alone does not provide complete information about a system or population.

Key Concept

1. Prior distribution:

⇒ This is initial probability distribution representing beliefs about the data before seeing any new data. based on prior knowledge.

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2. Posterior distribution:

⟹ This is the updated distribution after taking the new data into account.

3. Bayes theorem:

⟹ Also known as Baye's rule or bayes law.

⟹ Which is used to determine the probability of hypothesis with prior knowledge.

⟹ Depends on the conditional probability formula:

$$\therefore P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where,

- $P(A|B)$ is Posterior probability:
  - Probability of hypothesis A given the data B.
- $P(B|A)$ is likelihood probability:
  - Probability of the data B given the hypothesis A is true.

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- $P(A)$ is prior probability:
  - Initial probability of ~~data~~ the hypothesis A before seeing the data.
- $P(B)$ is marginal probability:
- is the evidence.
- The total probability of the data X.

Naive Bayesian classifier

It is a probabilistic ML model that is used for classification tasks.

Def:

⟹ it is Simple but most powerful classification technique.

⟹ Based on Bayes theorem.

⟹ it assumes that the attributes of a data sample are conditionally independent given the class.

⟹ Supervised learning algorithm.

⟹ it is a probabilistic classifier.

why is it called Naive bayes?

⟹ Comprised of two words

- Naive
- Bayes

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Naive:

⇒ Because it assumes that the occurrence of a certain feature is independent of the occurrence of other features.

Bayes:

⇒ Because it depends on the principle of baye's theorem.

Bayesian belief networks

⇒ BBNs provide a more sophisticated approach to modeling dependencies between variables.

⇒ They specify joint conditional probability distributions.

Components:

DAG:

- Each node represent variables which can be discrete or
- Continuous, Arc b/w them represent probabilistic dependencies.

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CPT:

- Each variable has CPT specifying its conditional distribution given its parents in the DAG.

Graphical model of causal relationships:

- BBNs provide a graphical model of causal relationships b/w variables.

Conditional independence:

- Each variable is conditionally independent of its nondescendents in the graph, given its parents.

Working of naive bayes classifier:

1. Define the hypothesis and data sample:
   - Let X be the data Sample whose label is unknown.
   - Let H be the hypothesis that X belongs to specific class C.
2. Calulate posterior probability

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3. Compute prior probability for each class:

$\Rightarrow P(C_i) = \frac{\text{number of training samples of class } C_i}{\text{total number of training samples}}$

4. Compute likelihood:

$$\Rightarrow P(X|C_i) = \prod_{t=1}^{n} P(x_i|C_i)$$

where x_i are the values of the attribute in the sample x.

5. Compute posterior probability:

- Use baye's theorem, the posterior probability is proportional to the product of the likelihood and the prior probability.

$$\Rightarrow P(C_i|X) \propto P(X|C_i) \cdot P(C_i)$$

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6. Class prediction:

- The class with the highest posterior probability is chosen as the predicted class for the samples X.

Pros:

- Fast & easy
- used for binary as well as multiclass classifications.
- For text classification is best.

Cons:

- it Assumes that all features are independent or unrelated. So, it cannot learn the relationship between features.

Applications:

- Used for credit scoring.
- medical data classification
- Spam filtering
- sentiment analysis

Example:

Sample	Attribute 1 A_1	Attribute 2 A_2	Attribute 3 A_3	Class C
1	1	2	1	1
2	0	0	1	1
3	2	1	2	2
4	1	2	1	2
5	0	1	2	1
6	2	2	2	2
7	1	0	1	1

1. Compute prior probabilities:

~~for C=1:~~

$P(C=1) = \frac{4}{7} = 0.5714$

$P(C=2) = 3/7 = 0.4286$

2. Compute Conditional Probabilities:

for C=1:

$P(A_1=1|C=1) = 2/4 = 0.50$

$P(A_2=2|C=1) = 1/4 = 0.25$

$P(A_3=2|C=1) = 1/4 = 0.25$

for C=2:

$P(A_1=1|C=2) = 1/3 = 0.33$

$P(A_2=2|C=2) = 2/3 = 0.66$

$P(A_3=2|C=2) = 2/3 = 0.66$

3. Compute posterior probabilities:

for C=1:

$P(X|C=1) = 0.50 \cdot 0.25 \cdot 0.25 = 0.03125$

$P(C=1|X) \propto 0.03125 \cdot 0.5714 - 0.0179$

for C=2:

$P(X|C=2) = 0.33 \cdot 0.66 \cdot 066 = 0.143$

$P(C=2|X) \propto 0.14375 \cdot 0.4286 = 0.0616$

4. class prediction:

- Bayesian inference, through the use of bayes theorem, allows for updating prior beliefs with new data to form posterior beliefs. The naive bayes classifier applies the principle to classification problems assuming conditional independence b/w attributes.

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Topic:

Predictive Regression

Regression analysis:

⟹ A statistical technique used to predict continuous values by determining the best model to relate output variable to one or more i/p variables.

Key components of regression analysis:

1. Dependent variable (Y)
2. Independent variable (X)

Objectives of regression analysis:

The primary goal of regression analysis is to find the best model that describes how the dependent variable Y is influenced by the independent variables. $x_1, \ldots, x_n$.

Goals of RA:

1. Cost-effectiveness predictions:
When measuring the inputs is cheap and measuring the o/p is expensive.

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2. Preemptive predictions:
   - When the i/p values are known before outputs, enabling early predictions.
3. Control and predictions:
   - Controlling i/p to predict the behavior of the o/p.
4. Causal relationships:
   - identifying causal r/s b/w i/p and o/p.

Interpolation vs regression:

interpolation:

⟹ Used when there is no noise in dataset. it aims to find a function $f(x)$ that exactly passes through all training points.

Regression:

⟹ Accounts for noise in the data. it aims to approximate the function $f(x_1, x_t)$ by a model $g(x_i)$, minimizing the empirical error.

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General linear regression model:

$$Y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

⇒ This equation shows that the output y is a linear combination of the input.

α: Intercept of regression line.
β: Slope of the regression line.

Linear regression with one variable:

$$Y = \alpha + \beta X$$

Least squares method:

- Used to solve for the coefficients α and β by minimizing the sum of squared errors.

$$SSE = \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2$$

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Calculating α and β:

- Compute the mean of X and Y.
- Use these means to calculate β:

$$\beta = \frac{\sum (x_i - mean_x)(y_i - mean_y)}{\sum (x_i - mean_y)^2}$$

3 Compute α:

$$\alpha = mean_y - \beta \cdot mean_x$$

Topic:
ANOVA

Q. Discuss the concept of ANOVA?

ANOVA: (Analysis of variance)

⇒ ANOVA is a statistical technique used to analyze the quality of a regression model and the influence of independent variables on the dependent variables.

⇒ Ronald Fisher introduced the concept of ANOVA.

⇒ it enables to test for significance of difference among more than two sample means.

⇒ Extension of T-test.

⇒ Given same result for comparing two samples.

Components of ANOVA

Total variation:

- Represented as the sum of squared deviations of each observation from the overall mean.

formula:

$$SS_{total} = \sum_{i=1}^{N} (Y_i - \bar{Y})^2$$

Between-Group variation:

⇒ The portion of the total variation that measures the variation between the group means. This reflect how much the group means differs from the overall mean.

Within-Group variation:

⇒ The portion of the total variation that measures the variation within each group.

formula:

$$SS_{between} = \sum_{j=1}^{k} n_j(\bar{Y}_j - \bar{Y})^2$$

$$SS_{within} = \sum_{j=1}^{k} \sum_{i=1}^{n_j} (Y_{ij} - \bar{Y}_j)^2$$

= Varian b/w sample + variance within sample

b/w

F-ratio test:

↳ Analysis of variance or variance ratio.

- F-ratio test is used in ANOVA.
- it is ratio of the mean between variation to the mean within-group variation.

$$= \frac{\text{Variance b/w}}{\text{Variance within}}$$

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Types of ANOVA

One-way ANOVA

- Used when comparing the means of three or more independent group based on one independent variable.

Two-way ANOVA

- used when comparing the means based on two independent variables, and can include interactions b/w the variable.

Assumptions about ANOVA:

- Samples follow normal distribution
- Samples have been selected randomly and independently.
- Each group should have common variance.
- Data are independent.

Null hypothesis:

$H_0: \mu_1 = \mu_2 = \dots, \mu_n$

means of all group are same.

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Alternate hypothesis:

- mean is different for atleast one group.

$H_a: \mu_1 \neq \mu_2 \neq \dots, \mu_n$

MANOVA

⇒ Stands for multivariate ANOVA. MANOVA generalize ANOVA for situations with multiple dependent variables, capturing correlations between them

multivariate linear model:

$$Y_j = \alpha + \beta_1 x_{j1} + \beta_2 x_{j2} + \dots + \beta_n x_{jn} + \epsilon_j$$

MANOVA test statistics:

- Roy's greatest root
- Lawley-hotelling trace
- Pillai trace
- wilks lambda

Applications:

- Experimental research
- Quality control
- Social science
- market research

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Topic:

Logistic Regression

⇒ it is popular machine learning algorithm under supervised learning algorithm.

⇒ used for predicting a categorical dependent variable using a set of independent variable.

Key concepts of logistic regression:

1. Categorical dependent variable:

- LR predicts the outcome of a categorical dependent variable.
- This means the output is discrete and can take categories such as Y/N, 0/1, T/F.

2. Probabilistic o/p:

- Does not provide exact binary outcomes (0 & 1).
- LR gives probabilistic values between 0 and 1, indicating the likelihood of the dependent variable being a specific category.

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3. Difference from linear regression:

- Linear regression used to predict continuous values while LR predict categorical output values.
- LR fits an "S"-shaped curve while linear regression fits a straight line.

4. logistic function:

- The logistic function is an "S" shaped curve that maps any real-valued number into a value between 0 and 1.
- The sigmoid function is:

$$\text{Sigmoid}(x) = \frac{1}{1+e^{-z}}$$

i/p function

Threshold value:

- Logistic regression uses a threshold value to classify the output. Typically if the probability is greater than 0.5, then o/p is classified as 1; otherwise, it is classified as 0.

Assumption of logistic regression:

- Dependent variable should be categorical.
- Independent variables should not be highly correlated with each other.

Logistic regression equation:

- Can be derived from linear regression equation by transforming it to ensure the output is between 0 and 1.
- Start with linear regression equation:

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

- Use the odds and logit transformation:

$$\frac{P}{1-P} = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n}$$

- Taking the logarithms

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$$

Types of logistic regression

- **Binomial**
  - Dependent variable has two possible outcomes.
- **multinomial**
  - Can have ~~three or more~~ possible unordered categories.
- **Ordinal**
  - Can have three or more ordered categories.

Calculating probabilities from logit:

$$P = \frac{e^{logit}}{1 + e^{logit}}$$

Example:

- Suppose logistic regression model:

$$\log\left(\frac{P}{1-P}\right) = -1.5 + 0.6X_1 + 0.3X_3 - 0.4X_2$$

- for a new sample input values $X_1 = 1,\ X_2 = 0\ \&\ X_3 = 1$

logit = ?

$$logit = -1.5 + 0.6(1) - 0.4(0) + 0.3(1) = -0.6$$

$$P = \frac{e^{-0.6}}{1 + e^{-0.6}} = \frac{0.5488}{1 + 0.5488} \approx 0.354 < 0.5 \quad \boxed{O/P = 0}$$

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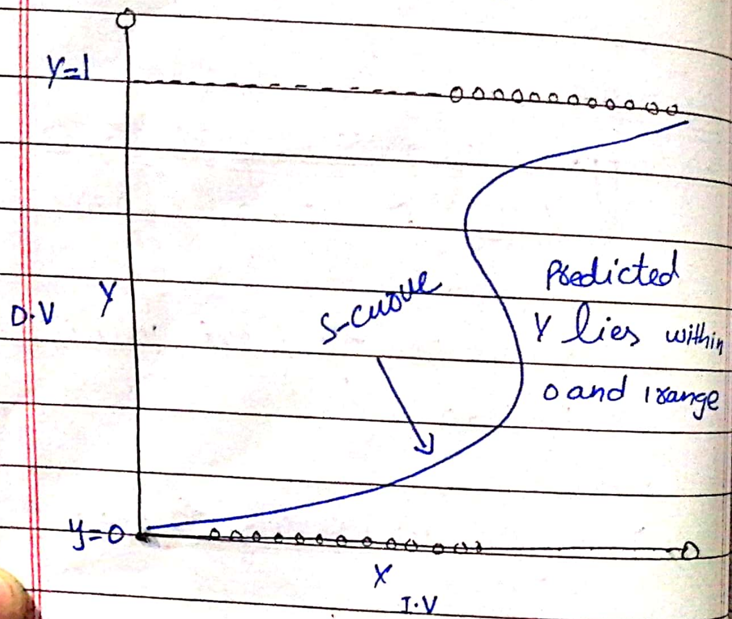
Applications:

- Probabilities and classification
- Versatality
- Effective variable identification.

Pros:

- Performs better when data is linearly separable.
- No require too many computational.



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Topic: Log linear model

Q. Introduce log-linear model using correspondence analysis of contingency tables.

Log-linear modeling: → to analyze categorical data. - how each category or value relates to all other categories or values in the dataset

- method used to analyze the relationships between categorical variables (sometimes quantitative).
- it approximates discrete, multidimensional probability distributions and is a type of generalized linear model.

Key Concepts

Aim: ⇒ To uncover any relationships b/w variables in order to make more reliable predictions

1. log linear model overviews
- The output Y_i is assumed to follow a poisson distribution with an expected value μ_j.
- The natural logarithm of μ_j is considered a linear function of the inputs:

$$\log(\mu_j) = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

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- This approach is used to identify associations between categorical inputs.

Categorical variables and contingency Tables:

- Categorical variables are represented in a frequency table, which shows the global distribution of data.
- Contingency table is a type of table in matrix format.
- Helps to examin relationship between two or more categorical variables.

NO specified output:

- Log linear modeling does not need one of the variables to be an output. used to analyze associations between all given categorical variables.

Correspondence analysis:

- Used to analyze the relationship b/w categorical

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variables using contingency tables.

Steps in log-linear model:

1. Transform the contingency table:

- Start by transforming the given contingency table into a table of expected values under the assumption that variables are independent.
- Expected values are calculated using row totals and column totals of the original tables.

2. Compare table using chi-square Test:

- Compare the observed values with the expected values using the chi-square test to determine if there is an association between the variables.
- The chi-square test statistic is calculated as:

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$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Example:

⇒ Consider a survey with 100 respondents examining support for abortion among males and females. Need to determine association b/w gender and support for abortion.
The contingency table:

Step-1: Create Contingency table

	Support Yes	Support No	Total
female	309	191	500
male	319	281	600
Total	628	472	1100

Step-2 calculate expected values:

$$E_{ij} = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

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calculations:

$$E_{11} = \frac{500 \times 628}{1100} = 285.5$$

$$E_{12} = \frac{500 \times 472}{1100} = 214.5$$

$$E_{21} = \frac{600 \times 628}{1100} = 342.5$$

$$E_{22} = \frac{600 \times 472}{1100} = 257.5$$

Now, the expected values table:

	Support Yes	Support No	Total
femal	285.5	214.5	500
male	342.5	257.5	600
Total	628	472	1100

Step-03 : Perform the chi-square test:

$$\chi^2 = \sum \frac{(O_i - E_{ij})^2}{E_{ij}}$$

$$\chi^2_{11} = \frac{(309 - 285.5)^2}{285.5} = 1.933$$

$$\chi^2_{12} = \frac{(191 - 214.5)^2}{214.5} = 2.574$$

$$\chi^2_{21} = \frac{(319-342.5)^2}{342.5} = 1.613$$

$$\chi^2_{22} = \frac{(281-257.5)^2}{257.5} = 2.161$$

Summing,

$$\chi^2 = 1.933 + 2.574 + 1.613 + 2.161$$
$$= 8.281$$

Step-04: Determine degree of freedom and critical value:

df = (no of rows − 1) x (no of col − 1)

$df = (2-1) \times (2-1) = 1$

- for a significance level (α) of 0.05, the critical value from the chi square table with 1 degree of freedom is 3.84.

Step-5 conclusion:

- $\chi^2 = 8.281$
- $\alpha = 0.05$ is 3.84

8.281 > 3.84, there is association b/w gender and support for abortion.

Topic: LDA

Linear discriminant analysis

Definition:

supervised dimensionality reduction technique

⇒ Used in classification problems where the goal is to predict the category of a dependent variable based on several independent variables. Also known as normal discriminant analysis.

⇒ It was originated from Fisher's linear discriminant.

Objective of LDA:

⇒ The main goal is to create a discriminant function that can distinguish between different classes. Maximize separability b/w 2 groups.

⇒ This function generates a score that is different for each class, helping to classify new data samples correctly.

- if we have two classes with multiple features and need to separate them efficiently then LDA used using single feature

The discriminant function:

⇒ A linear discriminant function can be written as:

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$Z = w_1x_1 + w_2x_2 + \dots + w_kx_k$

- $x_1, x_2, \dots, x_k$ are independent variables
- $w_1, w_2, \dots, w_k$ are the weights assigned to each independent variables.
- z is the discriminant score.

Constructing the discriminant function:

⇒ To construct the discriminant function, find a weight w_i that maximize the separation between classes. This is done by maximizing the ratio of between-class variance to within-class variance.

Between-class variance:

- measure how different the classes are from each other.

Within-class variance:

- measures how spread out the data points are within each class.

Criteria:

1) maximize the distance between means of two classes.
2) minimizes the variance within individual class

Using the discriminant functions

- Use to predict the class of new data samples.
- Compare the discriminant score z of the new sample to a cutoff value.

Cutting Scores:

⇒ A cutting score is a threshold that helps decide the class of a new sample

- if the classes of equal size and have the same variance, the optimal cutting score is the average of the mean discriminant scores of the two classes:

$$Z_{cut-ab} = \frac{Z_a + Z_b}{2}$$

mean discriminant scores

- if the classes are not of equal size, the optimal cutting score is weighted average:

$$Z_{cut-ab} = \frac{n_a \cdot Z_a + n_b \cdot Z_b}{n_a + n_b}$$

number of samples

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Classification rule:

⇒ A new sample is classified based on its discriminant score:

- if z > zcut-ab, the sample is classified into class one.
- if z < zcut-ab the sample is classified into the other class.

Properties of LDA:

⇒ Dimensionality reduction

⇒ Statistical properties

- mean
- covariance

⇒ multivariate Gaussian distribution

Assumptions of LDA:

- Gaussian distribution
  - each i/p data should have
- Linear separatability → classes
- equal covariance matrix → for all classes

Constraint:

- Not perform well in high dimensional space.

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Creating the LDA model:

1. Estimates the means and covariances.
2. Compute the discriminant function.
3. Predict new data points.

Multiple discriminant analysis:

⇒ Used for complex problems.

⇒ This term used in situations when separate discriminant functions are constructed for each class.

⇒ The classification rule is to assign the new sample to the class with the highest discriminant score.